
What does self-supervision add to engineered features for stellar light curves?

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Abstract

TESS light curves encode a star’s pulsation, rotation, binarity and oscillations, among other physical properties, but the survey produces far more light curves than can be labelled by hand. Astronomers have engineered a few dozen physically motivated features from them, which together remain a strong baseline. Previous work has trained encoders on light curves and judged them by their own downstream scores, but has not asked what they add to the engineered features. We pre-train a convolutional encoder on unlabelled TESS light curves to predict each window’s latent representation from those of its neighbours. We then freeze it, add its features to the 25 engineered features, and read the combination out with one fixed linear probe on ten tasks: five variability classes, four asteroseismic targets and rotation period. Adding the learned features improves the readout on 7 of 10 tasks over the engineered features alone, while an untrained encoder does not. The combination also outperforms two networks trained end-to-end on the same labelled stars on most tasks. An encoder trained without labels therefore gives astronomers new features that complement the ones they already use rather than replace them.

1 Introduction

A star’s light curve carries its pulsation, rotation, eclipses and oscillations, and with them its age, interior and companions. Wide-field surveys now produce light curves far faster than they can be labelled. TESS [Ricker et al., 2015] returns 2-minute or 20-second [Huber et al., 2022] light curves for hundreds of thousands of pre-selected stars, and full-frame images of every star in the field at cadences that change between mission cycles (30 minutes, then 10, now 200 seconds). LSST [Ivezić et al., 2019] will add *billions* more light curves, albeit at a cadence of days. Self-supervised learning is the natural response: pre-train one encoder on the unlabelled archive, then attach a cheap readout for each task. Encoders of this shape exist for ground-based, Gaia and Kepler light curves [Donoso-Oliva et al., 2025, Zuo et al., 2025, Rui, 2026] and are evaluated by classification accuracy, sometimes against engineered features [Rui, 2026]; a label-free TESS representation [Poznanski, 2026] is evaluated by whether repeat observations of one star lie close in its embedding space. None measures what they add to engineered features.

In this domain, the baseline that a learned encoder must improve on is already strong. Decades of variable-star research have distilled light curves into compact engineered summaries: periodogram peak structure, band-limited power, spectral entropy and robust scatter statistics. Pipelines built on them remain highly competitive. The benchmark of Li et al. [2025] makes the point sharply: across seven expert-labelled variable-star classes, no time-series foundation model they tested strictly surpassed the long-established domain feature baseline on classification. A learned encoder for this data therefore has to answer one question: do its features add predictive signal to the engineered features under the same readout?

We measure that directly. We pre-train an encoder on TESS light curves, freeze it, and add its features to the 25 engineered features under one fixed linear readout. We ask two things: what the learned features add, and whether the combination, which trains only a linear model on the labels, outperforms two networks trained end-to-end on the same labels. Our contributions are:

- **The learned features add to the engineered features rather than replacing them.** Adding the learned features to the engineered ones improves a fixed linear readout beyond 2 SE on 7 of 10 tasks; an untrained encoder gives no such gain, so the gain comes from pre-training, not from merely adding 128 columns.
- **The combined pipeline, read out by a fixed linear model, outperforms the two supervised networks we train end-to-end on the same labels on most tasks.** Against a supervised Conv1D with the encoder’s architecture it is ahead on 6 of 10 tasks and indistinguishable on the other 4; against an MLP on raw flux it is ahead on 9 of 10.

2 Data and method

Data. We use 2-minute SPOC PDCSAP light curves of stars brighter than TESS magnitude $T = 10$ from Sectors 27–101. Pre-training and five of the ten probes (pulsating, eclipsing binary, rotation, transit and rotation period) use a label-enriched subset of 13,470 stars, split 70/15/15 by star: every star labelled as a transit host, eclipsing binary or pulsator that has a usable window, plus a random sample of 10,000 quiet stars, listed in none of our variability catalogues. The pre-training loss never sees a label, but this pool is label-enriched, not a random draw from the survey. The other five probes draw their labels from catalogues that barely overlap this subset and are scored on a second pool of 22,860 stars built the same way (every catalogue positive with a usable window plus 10,000 quiet stars, split 70/15/15). No test star of the first pool enters pre-training. About 8% of the second pool’s test stars (265 of 3,429) were in the pre-training subset; removing them leaves all seven gains in place (Appendix F). Each light curve is split into contiguous segments wherever the time axis breaks, at a missing cadence or a jump longer than five median cadences; segments are median-normalised by their median absolute deviation and sliced into non-overlapping windows of 256 cadences ($256 \times 120 \text{ s} \approx 8.5 \text{ h}$); restricting to one cadence keeps every window the same span of time. Windows containing a missing value are discarded, never interpolated or padded, and segments are never stitched across sector or gap boundaries. Pre-training consumes sequences of exactly 16 consecutive windows ($\approx 5.69 \text{ d}$), cut at a random offset within a segment, and uses no labels.

Encoder. A 1D convolutional variational encoder [Kingma and Welling, 2014] maps each window to a 128-dimensional Gaussian posterior, and a GRU [Cho et al., 2014] predicts the next window’s latent from its predecessors, run both forward and backward over the sequence. The objective is

$$\mathcal{L} = \|\hat{x} - x\|^2 + \beta \text{KL} + \lambda [\|\hat{z}_{t+1} - \bar{z}_{t+1}\|^2 + \|\hat{z}_{t-1} - \bar{z}_{t-1}\|^2] \\ + w [\|\log P_H(\hat{x}) - \log P_H(x)\|^2 + \|\Delta\hat{x} - \Delta x\|^2],$$

where every term is a mean squared error, $\bar{z}$ the stop-gradient target of the forward and backward predictors, P_H the Hann-tapered power spectrum, Δ the first difference, and $(\beta, \lambda, w) = (0.1, 60, 0.3)$, β warmed up linearly over ten epochs. The encoder is trained with six seeds and then frozen; no gradient from any label reaches it.

Representations and readout. A star’s learned features are μ , the mean, over every window of its first segment (16 for 95% of stars, 20 for the rest), of the per-window posterior means. The engineered baseline is a fixed 25-feature vector modelled on the T’DA variability pipeline [Audenaert et al., 2021], computed on the same first segment, 19 from its periodogram and 6 time-domain statistics; Appendix E defines each feature. Readouts are fixed in advance and never tuned on a task, with inputs standardised using training-split statistics alone: class-balanced L_2 logistic regression at $C = 1$ for the detection tasks, ridge for the two regressions with its penalty set by leave-one-out cross-validation. We compare four arms, each the same readout on a different feature set: **features**, μ , **features** $\oplus \mu$, and, as a control, the same fusion built on an *untrained* encoder; the two supervised baselines below are the fifth and sixth arms. Appendix B repeats the first three under XGBoost.

Supervised baselines. Two end-to-end baselines see exactly the same input windows, star sets and splits. **C1** reuses the encoder’s convolutional trunk unchanged, adds a 128-unit linear layer and a

Table 1: Ten-task scorecard. Frozen arms: a fixed logistic or ridge readout, or XGBoost on the same inputs (Appendix B). Bold: best point estimate of the three arms under each frozen readout. n : test stars; prev.: positive-class prevalence.

task	metric	n (prev.)	frozen linear readout			frozen XGBoost readout			end-to-end supervised	
			features	μ	features $\oplus \mu$	features	μ	features $\oplus \mu$	Conv1D	MLP
pulsating	PR-AUC	2021 (0.107)	0.789	0.806	0.846	0.845	0.821	0.834	0.816	0.748
eclipsing binary	PR-AUC	2021 (0.097)	0.742	0.771	0.775	0.798	0.789	0.807	0.774	0.719
rotation	PR-AUC	2021 (0.089)	0.540	0.559	0.555	0.596	0.544	0.576	0.565	0.526
transit	PR-AUC	2021 (0.060)	0.190	0.144	0.218	0.217	0.223	0.228	0.134	0.109
oscillating giant	PR-AUC	3429 (0.383)	0.920	0.854	0.918	0.924	0.899	0.923	0.875	0.883
solar-like osc.	PR-AUC	3429 (0.122)	0.320	0.336	0.390	0.470	0.409	0.463	0.396	0.275
RGB vs. HeB	ROC-AUC	161 (0.702)	0.758	0.660	0.708	0.746	0.725	0.763	0.530	0.762
$\nu_{\max}$	R^2	1313	0.831	0.802	0.867	0.916	0.831	0.919	0.832	0.834
rotation period	R^2	150	0.703	0.677	0.717	0.695	0.655	0.730	0.597	0.698
flare activity	PR-AUC	3429 (0.089)	0.474	0.453	0.524	0.562	0.485	0.565	0.533	0.444

scalar output, and is trained per task with a class-weighted loss on a star-level score, the mean of the star’s window logits computed inside the network so that the gradient flows through the mean. **C2** replaces the convolutional trunk with a dense one and changes nothing else, so C1 versus C2 isolates the convolutional inductive bias. Both use one pre-registered regularisation setting for all tasks, select on a validation split no probe has ever touched, and are trained with six seeds.

Tasks and metrics. We score ten tasks with one probe per physical quantity: pulsating variables [Gao et al., 2025], eclipsing binaries [Prša et al., 2022], rotation and rotation period [Boyle et al., 2026], transits [Guerrero et al., 2021], flare activity [Seli et al., 2025, Vida et al., 2021], oscillating red giants and their $\nu_{\max}$ [Hon et al., 2021], solar-like oscillators [Hatt et al., 2023], and RGB versus helium-burning discrimination [Sreenivas et al., 2026]. Here $\nu_{\max}$ is the frequency of maximum oscillation power. Detection tasks are scored by the area under the precision–recall curve (PR-AUC), whose floor is the positive-class prevalence rather than 0.5; prevalence is therefore quoted with every detection score. The two regressions are scored by R^2 .

Uncertainties. All error bars are $2\times$ the standard error of the mean over seeds. For the μ arms the seeds are independent pre-training runs (six); for C1 and C2 they are initialisation and shuffling seeds (six); the engineered arm is deterministic. The seeds of the μ arms and of the supervised arms are unrelated, so differences between them are treated as unpaired and their bands combined as $\sqrt{\text{SE}_a^2 + \text{SE}_b^2}$. A difference counts only when it exceeds the combined 2 SE band. These bands measure re-training spread only; a paired bootstrap over the test stars (1,000 resamples, seeds averaged; Appendix D) checks that no verdict hinges on the particular test stars.

3 Results

The learned features add to the engineered features. Table 1 and Figure 1a give the comparison. Adding μ to the 25 features improves the fixed readout beyond 2 SE on 7 of 10 tasks, led by solar-like oscillators $+0.071$, pulsating $+0.057$ and flare activity $+0.051$, and reaching $+0.013$ (R^2) on rotation period. Rotation is positive but just short of the bar ($+0.016$ against 0.016). The two negative tasks are the ones where 128 extra columns cost more than they add: the 25 features already cover the task (oscillating giants, -0.003 on a baseline of 0.920) or the training set is too small to fit 128 more weights (RGB versus helium-burning, -0.050 on 755 training stars). On both, an untrained encoder loses by about the same amount as the trained fusion, within the error band; that is what redundant columns do to a linear fit, not a sign that μ carries misleading content (Appendix A). Across all ten tasks the untrained control is negative on seven and never exceeds $+0.016$ (Table 2), so the gains are not what any 128 extra columns would buy on this feature set; they are specific to what pre-training learned. μ on its own does not replace the features (ahead on only 4 of 10 tasks alone), as Li et al. [2025] also find for time-series foundation models; we therefore report fusion rather than substitution [Makinen et al., 2024].

The fusion probe outperforms the two supervised networks on most tasks. Against C1, which shares the convolutional trunk, labels and input windows but is trained end-to-end, the fusion probe

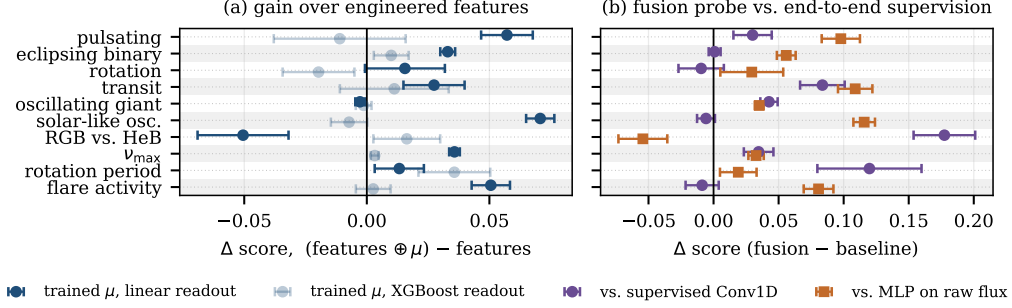


Figure 1: The two claims. **(a)** What the learned features add to the engineered features, under the fixed linear readout of Table 1 (solid) and under an XGBoost readout on the same inputs (faded; Appendix B). **(b)** The fusion probe against the two supervised baselines. Bars are 2 SE.

is ahead on 6 of 10 tasks and indistinguishable from it on the other 4 (Figure 1b). Against C2, an MLP on raw flux, it is ahead on 9 of 10 and behind on one (RGB versus helium-burning, 161 stars). C1 is not weak: it is ahead of the engineered arm on 5 of 10. A paired star-bootstrap (Appendix D) excludes zero on 5 of the 7 fusion gains, 5 of the 6 leads over C1 and 7 of the 9 over C2. This holds for this architecture on 669 to 16,002 labelled stars per task, not for supervision in general.

The GRU prediction term raises the gain on every task. Removing the GRU prediction term, all else fixed and the same six seeds, reduces the fusion gain on all 10 tasks, and by more than 2 SE on 6 when the two arms are differenced seed by seed (Table 2). Without it the gain falls by more than half on 5 of the 8 tasks where fusion gains.

4 Discussion

Pick the encoder by a held-out probe, not by its loss. Across the ten pre-training recipes of one sweep, a lower validation reconstruction loss went with a *lower* probe score: the Spearman ρ between negated loss and PR-AUC runs from -0.58 to -0.84 over four tasks. The recipe with the lowest loss is the one without the prediction term on all four, and choosing it would cost 0.021 – 0.049 PR-AUC (Appendix C). An encoder here therefore has to be chosen by a held-out probe, not by its own loss.

Limitations and next steps. *Readout.* The four linear-readout arms (features, μ , features $\oplus \mu$, and the untrained control) share one linear model. It has few parameters, so it cannot rescue weak features, and it is identical across the four, so the comparison is between feature sets. Our claim is confined to these 25 features and a linear readout: under XGBoost on the same inputs the gain exceeds 2 SE on 4 of 10 tasks and is negative on one (Appendix B). *Data.* Every arm, the supervised baselines included, sees one 5.7-day segment of a bright 2-minute star. The fainter, far more numerous stars in the full-frame images, and light curves spanning several sectors, are the obvious next experiments. *Labels.* Our negatives are stars absent from the label catalogues, not stars verified to lack the signal, so catalogue incompleteness adds positive-unlabelled label noise and the detection scores may be conservative. The flare label marks stars that have flared somewhere in their TESS data, not a flare inside the 5.7-day segment we encode, so the task is recognising a flaring star, not finding a flare. *Label scarcity.* When the readout is trained on a few percent of the labels, the advantage narrows on 7 of 10 tasks, so how much pre-training helps when labels are scarce remains open. *Discovery.* Every task here has a catalogue; whether μ also separates kinds of variability that no catalogue labels yet, for instance as clusters in the latent space, is untested.

Reproducibility and disclosure. §2 and Appendix E give the data, objective, features and readouts; the encoder has four convolutional stages (widths 32/64/128/256, kernel 5, each halving length), a 128-d latent and a one-layer GRU of width 256. Code and score tables will be released on publication; generative AI assisted with analysis code and prose editing.

References

- Jeroen Audenaert, James S. Kuszlewicz, Rasmus Handberg, et al. TESS data for asteroseismology (T²DA) stellar variability classification pipeline: Set-up and application to the Kepler Q9 data. *The Astronomical Journal*, 162(5):209, 2021. arXiv:2107.06301.
- Timothy R. Bedding, Benoit Mosser, Daniel Huber, et al. Gravity modes as a way to distinguish between hydrogen- and helium-burning red giant stars. *Nature*, 471:608, 2011. arXiv:1103.5805.
- Andrew W. Boyle, Luke G. Bouma, and Andrew W. Mann. The TESS all-sky rotation survey: Periods for 1,046,317 stars within 500 pc. *arXiv e-prints*, 2026. arXiv:2603.05586.
- Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, et al. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Conference on Empirical Methods in Natural Language Processing*, 2014. arXiv:1406.1078.
- Cristobal Donoso-Oliva, Ignacio Becker, Pavlos Protopapas, et al. Astromer 2. *arXiv e-prints*, 2025. arXiv:2502.02717.
- Xinyi Gao, Xiaodian Chen, Shu Wang, and Jifeng Liu. Classification of periodic variable stars from TESS. *The Astrophysical Journal Supplement Series*, 276(2):57, 2025. arXiv:2412.06175.
- Natalia M. Guerrero, Sara Seager, Chelsea X. Huang, et al. The TESS objects of interest catalog from the TESS prime mission. *The Astrophysical Journal Supplement Series*, 254(2):39, 2021. arXiv:2103.12538.
- Emily Hatt, Martin B. Nielsen, William J. Chaplin, et al. A catalogue of solar-like oscillators observed by TESS in 120-second and 20-second cadence. *Astronomy & Astrophysics*, 669:A67, 2023. arXiv:2210.09109.
- Marc Hon, Daniel Huber, James S. Kuszlewicz, et al. A ‘quick look’ at all-sky galactic archeology with TESS: 158,000 oscillating red giants from the MIT quick-look pipeline. *The Astrophysical Journal*, 919(2):131, 2021. arXiv:2108.01241.
- Daniel Huber, Timothy R. White, Travis S. Metcalfe, et al. A 20-second cadence view of solar-type stars and their planets with TESS: Asteroseismology of solar analogs and a re-characterization of π Men c. *The Astronomical Journal*, 163(2):79, 2022. arXiv:2108.09109.
- Željko Ivezić, Steven M. Kahn, J. Anthony Tyson, et al. LSST: From science drivers to reference design and anticipated data products. *The Astrophysical Journal*, 873(2):111, 2019. arXiv:0805.2366.
- Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. In *International Conference on Learning Representations*, 2014. arXiv:1312.6114.
- Weijian Li, Hong-Yu Chen, Nabeel Rehemtulla, et al. StarEmbed: Benchmarking time series foundation models on astronomical observations of variable stars. *arXiv e-prints*, 2025. arXiv:2510.06200.
- T. Lucas Makinen, Ce Sui, Benjamin D. Wandelt, et al. Hybrid summary statistics. *arXiv e-prints*, 2024. arXiv:2410.07548.
- Dovi Poznanski. A useful representation of TESS light curves. *arXiv e-prints*, 2026. arXiv:2605.05324.
- Andrej Prša, Angela Kochoska, Kyle E. Conroy, et al. TESS eclipsing binary stars. I. short-cadence observations of 4584 eclipsing binaries in sectors 1–26. *The Astrophysical Journal Supplement Series*, 258(1):16, 2022. arXiv:2110.13382.
- George R. Ricker, Joshua N. Winn, Roland Vanderspek, et al. The transiting exoplanet survey satellite. *Journal of Astronomical Telescopes, Instruments, and Systems*, 1(1):014003, 2015. arXiv:1406.0151.
- Yicheng Rui. Domain-informed multi-view self-distillation for astronomical light-curve representation learning with JEPA. *arXiv e-prints*, 2026. arXiv:2606.28446.

- Bálint Seli, Krisztián Vida, Katalin Oláh, et al. Stellar flare morphology with TESS across the main sequence. *Astronomy & Astrophysics*, 694:A161, 2025. arXiv:2412.12989.
- K. R. Sreenivas et al. Global asteroseismology of 19,000 red giants in the TESS continuous viewing zones. *arXiv e-prints*, 2026. arXiv:2604.00498.
- Krisztián Vida, Attila Bódi, Tamás Szklénár, and Bálint Seli. Finding flares in Kepler and TESS data with recurrent deep neural networks. *Astronomy & Astrophysics*, 652:A107, 2021. arXiv:2105.11485.
- Xiaoxiong Zuo, Yihan Tao, Yang Huang, et al. FALCO: A foundation model of astronomical light curves for time-domain astronomy. *arXiv e-prints*, 2025. arXiv:2504.20290.

A The two tasks where fusion does not help

task	features	fusion Δ	2 SE	untrained Δ	n_{test}
oscillating giants	0.920	−0.003	0.002	−0.003	3,429
RGB vs. He-burning	0.758	−0.050	0.019	−0.038	161

Oscillating giants: the 25 features already cover the task. The labels of Hon et al. [2021] come from a convolutional classifier reading a 128×128 image of each star’s power spectrum, which fires on a visible oscillation power excess at $6 \lesssim \nu_{\text{max}} \lesssim 250 \mu\text{Hz}$. Our engineered vector encodes close to that same quantity: eight log-spaced band powers spanning 0.3–360 c/d, plus the spectral centroid and entropy, locate the excess and describe its concentration. The features accordingly reach 0.920, and the trained and untrained fusion arms then lose the same 0.003: 128 collinear columns diluting a linear fit, with no trained-versus-untrained difference left to attribute to μ .

RGB versus helium-burning: no arm resolves the discriminant. Sreenivas et al. [2026] assign the evolutionary state with a convolutional classifier reading the background-corrected power spectrum *folded at $\Delta\nu$* ; the physics underneath is the dipole mixed-mode period spacing $\Delta\Pi_1$ [Bedding et al., 2011]. Neither is accessible at our baseline. The encoder sees one 256-cadence window at a time, so the finest frequency structure available to μ is $\Delta f = 32.6 \mu\text{Hz}$, while $\Delta\nu$ for these stars is $\approx 1.5\text{--}18 \mu\text{Hz}$; the comb itself is unresolved, let alone a fold of it. The engineered features, computed on the whole 5.69 d segment, reach $2.0 \mu\text{Hz}$, still coarser than the mixed-mode spacing $\Delta\Pi_1 \nu^2 \approx 0.2\text{--}0.6 \mu\text{Hz}$ near $\nu_{\text{max}} = 50 \mu\text{Hz}$. No arm on this row measures the quantity the label encodes; the 0.758 the features reach rides on correlates of evolutionary state rather than on its discriminant. The row behaves accordingly: it is the only one where C2 (0.762) is ahead of the fusion probe while C1 reaches only 0.530 on 755 training stars, and it carries the smallest test set in the study.

B The gain under a nonlinear readout

Table 2: What 128 added columns change, relative to the 25 features alone, under each frozen readout (mean over six encoder seeds; absolute scores are in Table 1). $+\mu$: our encoder. No prediction term: the same recipe trained without the GRU prediction term. Untrained: a randomly initialised encoder, which is a single encoder and carries no band. Bold: beyond 2 SE. The last row counts tasks won, tied and lost at that bar.

task	metric	linear readout			XGBoost readout	
		$+\mu$	$+\mu$, no prediction term	+untrained μ	$+\mu$	+untrained μ
pulsating	PR-AUC	+0.057	+0.051	+0.016	−0.011	−0.003
eclipsing binary	PR-AUC	+0.033	+0.012	−0.010	+0.010	−0.003
rotation	PR-AUC	+0.016	−0.011	−0.026	−0.020	−0.034
transit	PR-AUC	+0.027	+0.018	+0.004	+0.011	−0.027
oscillating giant	PR-AUC	−0.003	−0.006	−0.003	−0.001	−0.008
solar-like osc.	PR-AUC	+0.071	+0.024	−0.007	−0.007	−0.026
RGB vs. HeB	ROC-AUC	−0.050	−0.052	−0.038	+0.016	−0.031
$\nu_{\max}$	R^2	+0.036	+0.010	−0.001	+0.003	−0.005
rotation period	R^2	+0.013	−0.005	−0.057	+0.036	−0.004
flare activity	PR-AUC	+0.051	+0.033	+0.006	+0.003	+0.006
win / tie / loss		7 / 1 / 2			4 / 5 / 1	

The XGBoost readout takes the same inputs as the linear one. For each task and seed it is selected over a grid of maximum depth $\{3, 5\}$ and learning rate $\{0.05, 0.1\}$, with up to 1,000 rounds and early stopping after 50 rounds without improvement on a held-out 25% of the training stars; the test stars play no part in the selection. Under XGBoost the features-only arm also varies with the seed, through the validation split, so the gain is differenced seed by seed.

Under XGBoost the μ gain (features $\oplus \mu$ minus features alone) is smaller than under the linear readout on most tasks. XGBoost also scores higher than the linear readout on the 25 features alone on eight of ten tasks, and the two changes go together. On solar-like oscillators the features-only score rises from 0.320 to 0.470 and the μ gain falls from +0.071 to −0.007. On RGB versus helium-burning, where XGBoost does not raise the features-only score, the μ gain is larger under XGBoost. Across the ten tasks, the rise in the features-only score and the fall in the μ gain have a Spearman rank correlation of 0.92. This is consistent with part of the information μ supplies to a linear readout being already present in the 25 features, in a form only a nonlinear readout can use.

Table 2 also gives the same comparison with an untrained encoder in place of the trained one. Under XGBoost its 128 columns lower the score on nine of ten tasks and the trained μ scores above it on eight, so the gains that survive come from pre-training, not from adding columns. XGBoost on the 25 features alone also scores above the linear fusion probe on 7 of 10 tasks (Table 1). The body’s claim is about what μ adds under a fixed linear readout, not that the linear fusion probe is the strongest classifier available on these features.

C Validation reconstruction against downstream score

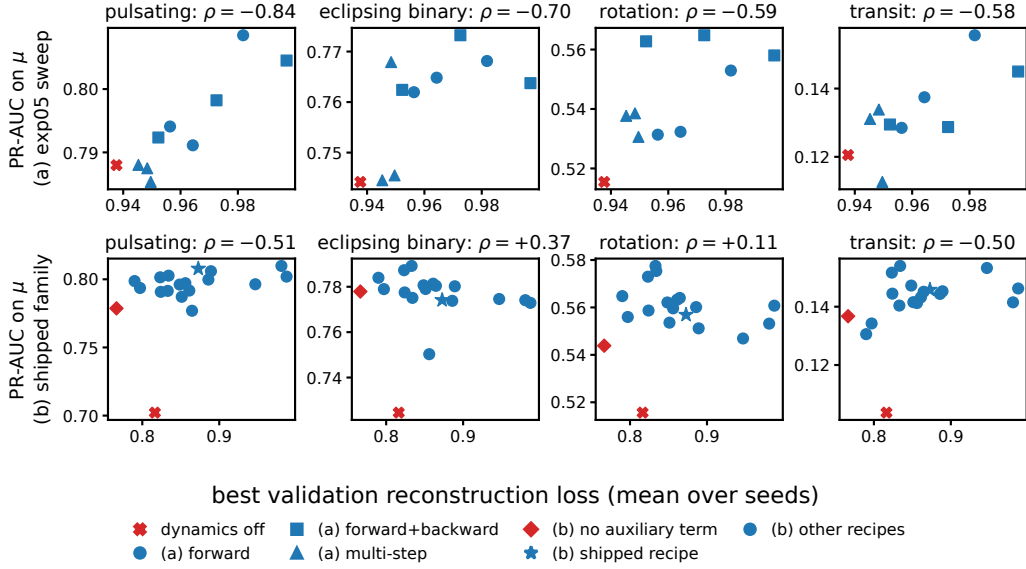


Figure 2: One point per pre-training recipe (mean over its seeds), one panel per task. (a) The exp05 sweep: ten recipes sharing corpus, architecture and loss terms, differing only in the GRU prediction term (off, forward, forward+backward, multi-step, the last three at three weights), four seeds each. (b) The family our encoder belongs to: 19 recipes on the final encoder that vary the form, weight and floor of the spectral auxiliary term, six seeds each; the diamond has no auxiliary term, the cross is the one dynamics-off recipe, the star is the recipe we use. x : best post-warm-up validation reconstruction loss; y : PR-AUC of the fixed linear probe on μ ; ρ : Spearman’s rank correlation between negated loss and score.

Figure 2 is the measurement behind the model-selection paragraph of §4. In row (a) the loss-optimal recipe is the dynamics-free one (red cross) on all four tasks, and it is the worst or next-to-worst probe on each, so selecting on the loss costs 0.021–0.049 PR-AUC against the probe-optimal recipe. The trend is not carried by that one point: with it removed, ρ still runs from -0.43 to -0.87 . Those recipes are an earlier sweep than our encoder (they carry an earlier spectral auxiliary term), so row (b) repeats the estimator on the family we ship: 19 recipes built on the final encoder that vary the form, weight and floor of the auxiliary term, six seeds each (two more recipes in which one seed collapsed are excluded; we never drop a seed). There the anti-correlation does not persist as such: ρ is -0.51 on pulsating and -0.50 on transit, $+0.37$ on eclipsing binaries and $+0.11$ on rotation. What persists is that the loss is not a selector: its optimum is again the recipe with the weakest competing term (the diamond, no auxiliary term at all), which is never the probe optimum, and trusting it costs 0.011–0.034 PR-AUC. So row (a) shows an anti-correlation on every task; in row (b) it survives on two tasks and reverses on two; in neither does the loss choose. Both are statements across recipes only: within one recipe, across its seeds, no label-free curve metric we tried (reconstruction, reconstruction plus auxiliary, active latent units) ranks the seeds, every such correlation sitting within ± 0.3 of zero.

D Paired bootstrap over test stars

The 2 SE bands in the body come from re-training and say nothing about whether the particular test stars favoured one arm. For every printed contrast we therefore resample each task’s test set with replacement 1,000 times, rescore every arm on the identical resampled list from its saved per-star predictions (no model is retrained), average each arm over its seeds, and take the difference. Table 3 gives the 95% percentile interval of that difference and the share of resamples in which the fusion arm is ahead; intervals that exclude zero are bold. The check tightens no verdict and reverses none:

every row the body calls ahead has the fusion arm ahead in at least 65% of resamples. It does loosen the smallest tasks – rotation period (150 test stars) and RGB versus helium-burning (161) account for four of the ten rows where a resolved 2 SE verdict becomes an interval that touches zero.

Table 3: Paired star-bootstrap of the four contrasts the body counts. Each entry is the 95% interval of (left arm – right arm) over 1,000 resamples of the test stars, seeds averaged, and the share of resamples with the difference above zero ($f_{>0}$; 1.00 means the left arm is ahead in every resample, 0.00 that it is behind in every resample). Bold: interval excludes zero.

task	fusion – features		fusion – C1		fusion – C2		fusion – dynamics-off fusion	
	95% CI	$f_{>0}$	95% CI	$f_{>0}$	95% CI	$f_{>0}$	95% CI	$f_{>0}$
pulsating	[+0.012, +0.105]	0.99	[−0.008, +0.073]	0.94	[+0.045, +0.150]	1.00	[−0.023, +0.036]	0.69
eclipsing binary	[+0.005, +0.062]	0.99	[−0.024, +0.025]	0.52	[+0.022, +0.094]	1.00	[+0.004, +0.038]	0.99
rotation	[−0.026, +0.057]	0.77	[−0.056, +0.040]	0.38	[−0.023, +0.079]	0.87	[−0.005, +0.052]	0.94
transit	[−0.025, +0.065]	0.85	[+0.041, +0.124]	1.00	[+0.046, +0.174]	1.00	[−0.016, +0.035]	0.78
oscillating giants	[−0.010, +0.005]	0.21	[+0.028, +0.057]	1.00	[+0.021, +0.051]	1.00	[−0.002, +0.008]	0.87
solar-like osc.	[+0.044, +0.097]	1.00	[−0.037, +0.022]	0.34	[+0.079, +0.151]	1.00	[+0.025, +0.067]	1.00
RGB vs. HeB	[−0.087, −0.012]	0.00	[+0.099, +0.247]	1.00	[−0.126, +0.020]	0.08	[−0.037, +0.039]	0.52
$\nu_{\max}$	[+0.028, +0.043]	1.00	[+0.024, +0.045]	1.00	[+0.018, +0.047]	1.00	[+0.020, +0.031]	1.00
rotation period	[−0.038, +0.067]	0.65	[+0.054, +0.186]	1.00	[−0.053, +0.100]	0.69	[−0.021, +0.060]	0.83
flare	[+0.019, +0.078]	1.00	[−0.047, +0.029]	0.32	[+0.027, +0.128]	1.00	[−0.000, +0.035]	0.97

E The 25 engineered features

All 25 features are computed from a star’s first segment: its windows concatenated into one flux series x of 4,096 or 5,120 cadences, already median-normalised by its median absolute deviation. The periodogram is the power spectrum of the mean-subtracted series, $P(f) = |\text{FFT}(x - \bar{x})|^2$ with the zero-frequency bin dropped, and $p(f) = P(f) / \sum_f P(f)$ its normalisation to unit sum. Frequencies are in cycles per day. f_1, f_2, f_3 are the three strongest local maxima of P , strongest first.

Table 4: The engineered feature vector, modelled on the T’DA variability pipeline [Audenaert et al., 2021].

#	feature	definition
1–3	$\log_{10} f_1, \log_{10} f_2, \log_{10} f_3$	frequencies of the three strongest periodogram peaks
4–6	p_1, p_2, p_3	normalised power at each peak, $p_k = P(f_k) / \sum_f P(f)$
7–8	$p_2/p_1, p_3/p_1$	peak-power ratios
9	harmonic ratio	$P(2f_1)/P(f_1)$ at the bin nearest $2f_1$; 0 if $2f_1$ exceeds the Nyquist frequency
10–17	band powers	$\log_{10} \sum_{f \in B_j} p(f)$ for eight log-spaced bands B_j spanning 0.3–360 c/d
18	spectral entropy	$-\sum_f p(f) \ln p(f) / \ln N$, N the number of frequency bins
19	spectral centroid	$\log_{10} \sum_f f p(f)$
20–21	skew, kurtosis	sample skewness and excess kurtosis of x
22	point-to-point scatter ratio	$\text{std}(\Delta x) / \text{std}(x)$, Δ the first difference
23	depth	95th minus 5th percentile of x
24	MAD	median absolute deviation of x
25	IQR	inter-quartile range of x

F Evaluation restricted to stars never seen in pre-training

The encoder is fitted on the training stars of the first pool and monitored on its validation stars, so the test stars of the five tasks scored on that pool are unseen by construction. The second pool shares some stars with it. Table 5 removes every test star that was in the pre-training training or validation split and rescores the saved per-star predictions; no model is refitted. The gain exceeds 2 SE on the same 7 of 10 tasks; the largest change is flare activity, +0.051 to +0.041, and the small loss on oscillating giants (−0.003) becomes a tie (−0.001).

Table 5: The fusion gain (features $\oplus \mu$ minus features, linear readout) on all test stars and on the test stars never seen in pre-training. Bold: beyond 2 SE over six encoder seeds.

task	metric	test stars	unseen test stars	gain, all	gain, unseen
pulsating	PR-AUC	2021	2021	+0.057	+0.057
eclipsing binary	PR-AUC	2021	2021	+0.033	+0.033
rotation	PR-AUC	2021	2021	+0.016	+0.016
transit	PR-AUC	2021	2021	+0.027	+0.027
oscillating giant	PR-AUC	3429	3164	−0.003	−0.001
solar-like osc.	PR-AUC	3429	3164	+0.071	+0.069
RGB vs. HeB	ROC-AUC	161	149	−0.050	−0.048
$\nu_{\max}$	R^2	1313	1218	+0.036	+0.034
rotation period	R^2	150	150	+0.013	+0.013
flare activity	PR-AUC	3429	3164	+0.051	+0.041